

Name: _____

Class: _____

Date: _____

Lab 7: Data Detective

Python Lab · Section 2.3–2.4 Data



Goal

Use a program to clean, filter, and summarize a data set and discover patterns.

Warm-up: Predict

```
temps = [72, 68, 75, 80]
print(len(temps))
print(max(temps))
print(sum(temps) / len(temps))
```

Predict the output.

AP Connection

- Programs filter, sort, and transform data (2.4).
- Correlation is not causation (2.3).

Build It

Step 1. Here is survey data: hours of sleep and test scores for 10 students.

```
sleep = [8, 6, 7, 5, 9, 6, 8, 4, 7, 9]
scores = [88, 72, 85, 65, 92, 70, 90, 60, 80, 95]
```

Step 2. Summarize: print the average, max, and min of each list.

```
print("Avg sleep:", sum(sleep) / len(sleep))
```

Step 3. Filter: print the scores of students who slept 8+ hours.

```
for i in range(len(sleep)):
    if sleep[i] >= 8:
        print(scores[i])
```

Step 4. Clean: this list of states is messy. Make every entry uniform.

```
states = ["CA", "Calif.", "california", "TX", "texas"]
# write code (or rules) to change them all to "CA" or "TX"
```

Step 5. Pattern: Do students who sleep more score higher? Is that causation? Write a sentence.

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Build It (continued) · Level Up

Level Up Challenges

Mild Sort the scores with `sorted()` and print the top 3.

Medium Print a text bar chart: one * for every 10 points of each score.

Spicy Find the average score for students who slept under 7 hours vs. 7 or more.

Python → AP Pseudocode

```
for s in scores:
    if s >= 90:
        print(s)
```

Exit Ticket

Name one question this data CAN answer and one it CANNOT.